Abstract

The invention discloses a system and method for local right confirmation and incremental synchronization of cognitive assets based on hash fingerprints. The system includes a Local Security Zone, a Distributed Incremental Right Confirmation Ledger and a Computing Power Isolation Operating Environment. The Local Security Zone slices a local plaintext knowledge base into semantic blocks and generates feature vectors in an offline state, then performs cryptographic coupling with signature salt values generated by a local private key to generate globally unique unidirectionally encrypted hash fingerprints. Binary bit sequences of the fingerprints can be used as seeds to covertly embed grammatical deviation watermarks, and original asset bodies are stored locally in an encrypted manner. The Distributed Incremental Right Confirmation Ledger receives and records incremental hash fingerprints and timestamps for ownership evidence storage through time sequence verification, without storing any plaintexts or vector indexes. When a session call occurs, the Computing Power Isolation Operating Environment applies for a temporary read-only sandbox to load Top-K plaintext fragments and binds a lifecycle clock. When the session ends or the clock resets to zero, the degaussing module performs multi-pass overwriting on sandbox memory by sequentially writing all-zero values, all-one values and random bit noise, and forcibly sends TCP RST packets to erase transmission traces, so as to realize controlled calling with zero residue.

Figure 1 is designated as the accompanying drawing of the abstract.